Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering

B.Tech. II (CSE) Semester-III
Database Management Systems (CS203)

Academic Year: 2026-27

Database Project

Design, develop, and demonstrate the database application with **your selected and registered topic**. There will be 2 to 3 students in a project's team. Select TEAM NAME for your project. Submit the required details in the provided **B. Tech. II CSE DBMS Project Google Sheet.** The instructions for preparing the database project documentation are provided in the attached Google document, and the submission link will be provided for the final project submission.

You will develop a project and maintain a Project Report that includes a section for each of the following phases:

A. Project Requirements

Analysis of the requirements of the project and a high-level description of the tasks involved.

B. Conceptual Design

This stage involves the following :

- Develop an Entity-Relationship model detailing the relations involved.
- Identify a minimal set of attributes that identify the entities and the relations.
- Mention cardinalities.
- List the constraints for each relation and entity.
- You should be able to explain the reasons for the particular design approach you have chosen.

C. Logical Design

This stage involves the mapping of your conceptual design above onto the relational data model.

- Design the tables for all your entities and relations.
- You should apply all normalizations you find useful and/or necessary.
- Ensure that your design still satisfies the user requirements. Justify your design choices.

D. Physical Design

This stage involves the following:

- Design a database based on the design developed in section B.
- Implement tables for the relations with the appropriate constraints.
- Maintain scripts for table creation, data insertion, modification, and deletion.
- Create indices for the database application. Justify the reasons.
- Justify your database design choices.

E. Prototype, Development, and Testing

This consists of the following:

- Design web interfaces for your application to meet user requirements.
- Write web interface code to access, update, and administer the SQL tables made.

F. Project Presentation

- Prepare a presentation and demonstrate your project, which will be scheduled before the end of the semester's practical exam week.